

Predicting Weekly Dengue Cases from Climate History

DengAI competition – final project report

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Abstract

Dengue is a mosquito-borne disease whose weekly case counts depend on weather that happened weeks or months earlier. In the DengAI competition we forecast weekly cases for San Juan (Puerto Rico) and Iquitos (Peru) from climate and vegetation measurements. This report is the record of an iterative investigation rather than a description of one final model. XGBoost and CatBoost first looked stronger than a forest on training and chronological validation, yet the public leaderboard reversed that ranking. That failure led us to a lower-variance, city-specific 500-tree system with at least five samples per leaf (22.5 public leaderboard MAE), and then to the more important discovery that all tree models were missing outbreak peaks. We traced the problem to delayed climate effects, selected feature- and city-specific history windows, and compressed the 461-input San Juan history into 180 stable multiscale summaries. The final submission combines that San Juan model with the preserved 380-input Iquitos raw-history model and round-to-nearest post-processing. Public leaderboard MAE improved from 26.7 to 22.5, 18.8, 16.6, and finally **15.9832**. Exact reproduction of all 416 submitted predictions and their SHA-256 hash establishes which recipe produced the score. Our results highlight the importance of temporal feature engineering: compact, multiscale climate representations improved the performance of a fixed MLP architecture, while city-specific model selection preserved the representation that worked best for each city. These findings suggest that designing informative input representations can be more effective than increasing network complexity for small climate-based prediction datasets.

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1 Problem and motivation

Dengue fever is transmitted by *Aedes* mosquitoes. Temperature, humidity and rainfall influence mosquito breeding, survival and the incubation of the virus, but each with a delay: a wet period now tends to translate into more infections several weeks later. Public-health agencies must decide *in advance* where to place vector-control teams, hospital beds and diagnostic supplies, so a forecast of next weeks' case counts that is available before the cases appear has direct operational value: it can shift effort from reaction to prevention.

The DengAI competition [1] frames this as a supervised regression problem. For each of two cities we are given weekly records of satellite vegetation indices (NDVI), station temperature and rainfall, and reanalysis climate variables, together with the number of reported cases (`total_cases`) for the training period. The task is to predict `total_cases` for a later, hidden period and the score is the mean absolute error

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{g}_i|. \quad (1)$$

What makes the task hard.

- The two cities have very different climates and case scales, so one model may not fit both.
- Case counts are heavy-tailed: most weeks are quiet, a few weeks are large outbreaks, and absolute error is dominated by how well those peaks are anticipated.
- Weather affects cases with city- and variable-specific delays, so the way the past is presented to a model matters.
- Training data are small (a few hundred to about a thousand weeks per city) and time-ordered, so naive validation is optimistic.
- True case counts for the test period are unavailable, so *previous cases cannot be used as predictors*; the model must forecast from climate history alone.

Executive Summary. Our work was a sequence of failures and questions, each created by the previous result (Figure 1). Boosting won locally but lost on the hidden period, so we reduced tree variance. The resulting forest still missed outbreak peaks, so we stopped tuning the learner and inspected the time series. That investigation suggested delayed infection-cycle effects, which led to feature-specific climate histories. Those histories helped but were too large for only 1456 rows, which led to multiscale compression. The final step was disciplined selection and verification: keep multiscale summaries for San Juan, retain raw histories for Iquitos, use the validated rounding rule, and archive the exact file and software environment.

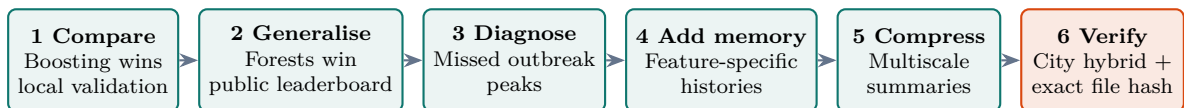


Figure 1: The experiment map. A disappointing result or visible failure mode triggered each next experiment.

2 Data

2.1 Loading and splitting

We load the three CSV files with **pandas**: features, labels, and the test features. Training features and labels are joined on the key (**city**, **year**, **weekofyear**) with a one-to-one check, giving a table with 1456 rows. The inputs X are the weekly climate columns and the target y is **total_cases**. The test keys are verified to match the submission template exactly, so predictions can be written in the required order.

Table 1: Dataset overview.

	San Juan (SJ)	Iquitos (IQ)
Training weeks	936	520
Test weeks (public)	260	156
Training period	1990–2008	2000–2010
Median weekly cases	19	5
Mean weekly cases	34.2	7.6
Maximum weekly cases	461	116

The 24 test columns comprise three key columns, the date, four NDVI indices, and 16 weather series from stations and reanalysis. Tree models use NDVI summaries as well as weather. The final MLP expands only the 16 consistently defined weather series, plus calendar phase: NDVI has substantially more missingness and did not provide stable lag-window evidence. This is why “24 columns” becomes 16 temporally engineered signals rather than 24 lagged variables.

2.2 Two cities, two target distributions

Figure 2 shows the weekly case series and their distributions. San Juan has rare, very large outbreaks (maximum 461, median 19), while Iquitos has smaller counts with many near-zero weeks (maximum 116, median 5). Both distributions are strongly right-skewed.

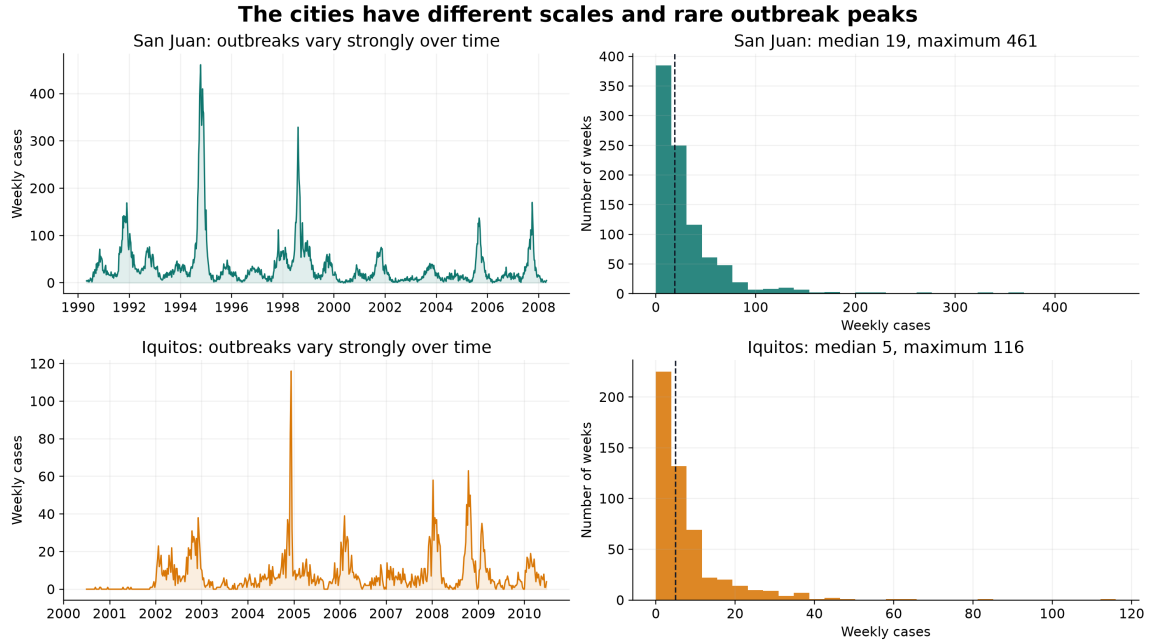


Figure 2: Weekly cases per city (left) and their distributions with the median marked (right). The scales differ by roughly a factor of four and both are dominated by rare peaks.

Decision. The cities differ in scale, climate and outbreak behaviour, so we train and tune a *separate model for each city*. Because the metric is absolute error and outbreaks are rare, we also report error separately on outbreak weeks (weeks above a city-specific high-case threshold) in addition to the mean.

3 Exploratory analysis

3.1 Seasonality helps but does not decide

Figure 3 plots the mean, median and interquartile range of cases per calendar week. San Juan peaks in the second half of the year (highest mean around week 40) and Iquitos shows a late-year peak (around week 50) with an additional early-year bump. However, the interquartile bands are wide: the same week of the year can have very different outcomes in different years.

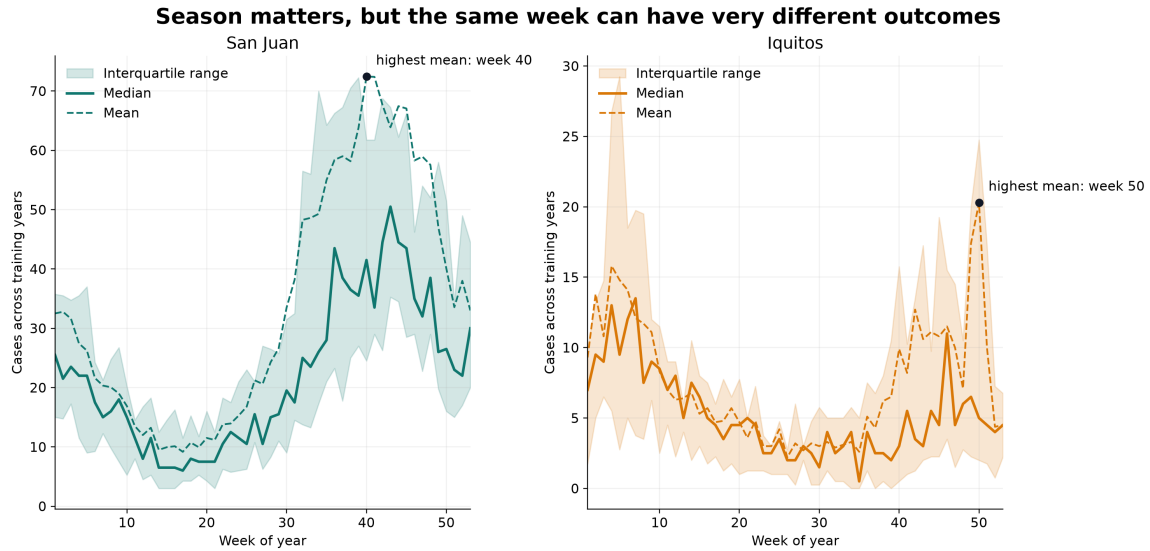


Figure 3: Cases by calendar week across training years. The seasonal signal is real, but the wide band shows that calendar week alone does not determine the count.

Decision. The model needs the *calendar phase* (as continuous sine/cosine terms so that week 52 and week 1 are neighbours) *and* recent climate conditions.

3.2 Missing values

Missing values are rare but not negligible: the training features contain 548 missing cells. Figure 4 lists the variables missing in at least 1% of training weeks. The northeast NDVI index in San Juan is the worst case (about 20%), followed by several Iquitos station temperature variables (about 7%).

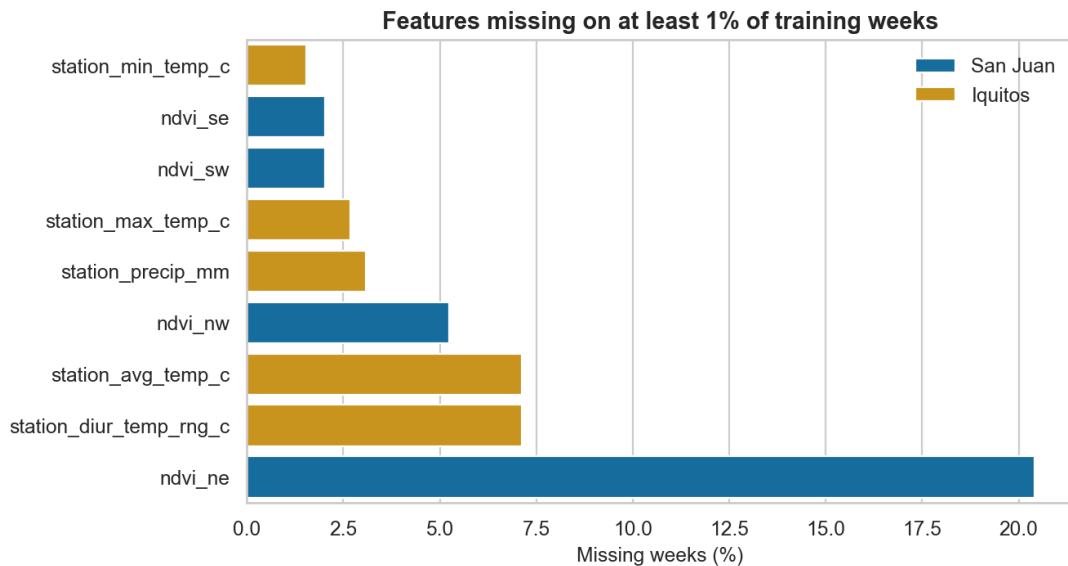


Figure 4: Share of training weeks with a missing value, by variable and city.

3.3 Case counts have memory; climate memory is variable-specific

Figure 5 illustrates two important characteristics of dengue incidence and climate history.

First, the autocorrelation of weekly dengue cases decreases gradually over approximately 15–20 weeks, with stronger temporal persistence in San Juan than in Iquitos.

This suggests that dengue incidence is not independent across weeks and motivates investigating historical climate information.

Second, different weather variables appear to require different history lengths.

To investigate this, we considered candidate history windows ranging from 3 to 51 weeks for each weather variable and city.

For a variable v , a window of length L contains consecutive observations:

$$\mathbf{x}_t^{(v,L)} = [x_{t-L+1}^{(v)}, \dots, x_t^{(v)}]. \quad (2)$$

A decision-tree screening procedure was used to compare candidate window lengths.

However, the original screening employed ordinary ten-fold validation rather than strictly chronological folds. Therefore, the selected lengths should be interpreted as exploratory feature-engineering choices, not unbiased estimates of optimal forecasting horizons.

The results suggest that useful climate history differs across variables and cities.

For example, the selected relative-humidity history was 34 weeks for San Juan but only 7 weeks for Iquitos.

These observations motivated the city-specific historical representations investigated in Section 6.3.

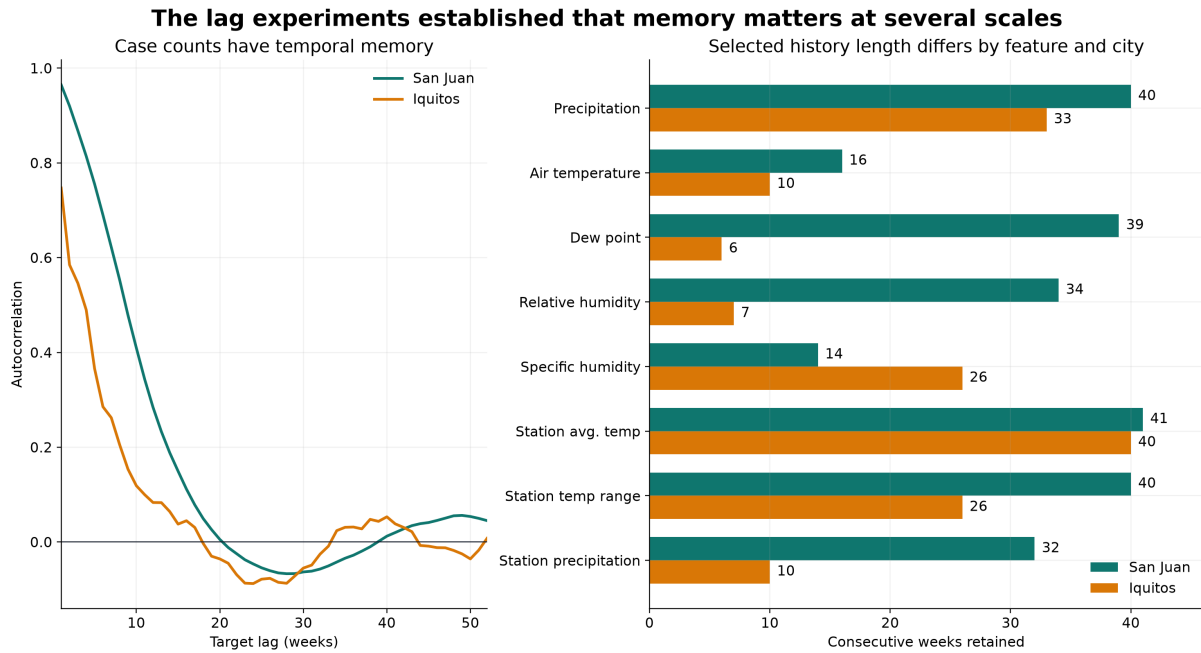


Figure 5: Left: autocorrelation of cases. Right: history length (number of consecutive weeks retained) chosen by the screening step, per weather variable and city.

Caveat. The original ten-fold procedure used for this screen was not fully forecasting-safe, so exact values such as 39 versus 40 weeks should not be read as biological constants. We use the screen only as evidence that memory is long, variable-specific and city-specific.

4 Preprocessing and feature engineering

Every transformation below is motivated by one of the findings above. Test-time features use only climate observations available at the model’s recorded history endpoint. For exact experimental compatibility, the reproduction notebook also retains the original two-row alignment and circular context used to construct the earliest training windows; we document this implementation quirk rather than silently changing the submitted model.

Table 2: Preprocessing steps and the reason for each.

Step	What we do	Why
Missing values, forests	Median imputer fitted on the training fold	The median is robust and fold-fitting avoids leakage
Missing values, boosting	XGBoost/CatBoost native missing-value routing	Each split learns a default direction for unavailable values; no global fill value is required
Missing values, MLP	Linear interpolation in time order	Weather is smooth over a week; interpolation avoids breaking history windows
Scaling (tree models)	None: raw units are retained	Split order is unchanged by monotone rescaling; scaling adds no benefit to XGBoost, CatBoost, Random Forest or Extra Trees
Scaling (MLP)	z -scores of weather columns using San Juan statistics; week of year min-max scaled	Variables are in different units; networks are scale-sensitive
Season	$\sin / \cos$ of $2\pi \text{ week}/52$ (and 4π for tree models, four harmonics)	Smooth, continuous yearly cycle across the new-year boundary
Target	Predict cases directly with an absolute-error loss; Iquitos trees use $\log(1+y)$	MAE is the competition metric; log stabilises Iquitos’ small counts for forests
Per-city models	One model and one set of hyper-parameters per city	Different scales, histories and regimes (Section 3)

4.1 Exactly what XGBoost and CatBoost received

The boosting comparison used the same information as the confirmed forest, so it was a model comparison rather than a different-feature comparison. We sorted each city chronologically, removed identifiers that should not drive prediction (`city`, date and target), and removed calendar year for the boosting candidates because extrapolating a numerical year trend into a three- or five-year test block was unsafe. All remaining inputs were numeric; CatBoost was not given categorical columns.

For each weekly row, the design matrix contained current climate and NDVI measurements, eight Fourier coordinates (four annual harmonics), NDVI mean and spatial spread, two temperature ranges, a temperature–humidity interaction, nine weather variables at lags 1, 2, 4, 8, 12, 16, and past-only means over 2, 4, 8, 12, 16 weeks. Rolling means were shifted by one week. Consequently, “past mean 4” at week t used $t - 1$ through $t - 4$, never the current value and never a future value. Previous dengue counts were excluded because true case labels are unavailable throughout the hidden horizon.

Table 3: Preprocessing and regularisation in the first model comparison.

Model	Input handling	Main controls used in the experiment
XGBoost	Same engineered numeric matrix; missing values handled by the learner; no standardisation	500 boosting rounds, learning rate 0.03, depth 2 or 3, minimum child weight 10, row subsampling 0.85, feature subsampling 0.8, L1 0.1, L2 10; MAE and Poisson objectives tested
CatBoost	Same engineered numeric matrix; native missing-value routing; no categorical features and no standardisation	500 iterations, learning rate 0.03, depth 4 or 6, L2 leaf penalty 10, Bernoulli subsampling 0.85, random strength 0.5; MAE and Poisson losses tested
Random Forest / Extra Trees	Fold-fitted median imputation; no standardisation	500 trees and at least five samples per leaf; city-specific target/criterion described in Section 6.2

The boosting targets were the observed case counts: Poisson candidates enforced a count-compatible objective and MAE candidates matched the leaderboard loss. We did not normalize the target. Only the Iquitos forest applied $\log(1 + y)$ and transformed its predictions back with $\exp(\hat{z}) - 1$. Every learned imputation rule and every fitted model used only the earlier part of each chronological fold.

4.2 Three feature sets

(A) Tree features. For each week: current climate; four annual Fourier harmonics (eight seasonal coordinates); NDVI mean and spread over the four locations; temperature ranges; a temperature $\times$ humidity interaction; climate at lags 1, 2, 4, 8, 12, 16; and trailing means over 2, 4, 8, 12, 16 weeks (computed with a one-week shift so the current row cannot leak into its own summary).

(B) Raw lag windows. For each weather variable v and city c we keep a window of $L_{v,c}$ consecutive past values (not a single lag). Concatenating all windows gives one flat vector: 461 inputs for San Juan and 380 for Iquitos. A 40-week window keeps all 40 consecutive values, not only the value at lag 40.

(C) Multiscale summaries. For each of the 16 weather variables we compute 11 statistics from the previous 52 weeks, and add four seasonal terms:

1. the current (latest) value;
2. means over 2, 4, 8, 13, 26, 52 weeks (climate level at several timescales);
3. standard deviations over 4 and 13 weeks (recent and medium-term variability);
4. the 4-week mean minus the 13-week mean (short-term conditions relative to baseline);
5. a 13-week trend, $(x_t - x_{t-12})/12$ (direction and rate of change);
6. sin, cos of 2π week/52 and of 4π week/52 (cyclic seasonality).

The size is $16 \times 11 + 4 = \mathbf{180}$ inputs. The final San Juan model uses these 180 inputs; the final Iquitos model retains its 380 raw-lag inputs.

5 Validation protocol and metric

5.1 Why chronological validation

A random split places neighbouring outbreak weeks on both sides of the split. The model can then “interpolate” an outbreak from adjacent weeks that would never be available in a real forecast, giving an optimistic score. The public test period lies strictly after the training period, so validation must have the same direction.

We use **expanding-window folds**: fold k trains on all weeks before a cut point and is evaluated on the next 52 weeks (Figure 6). All preprocessing that learns from data (imputation, scaling, any projection) is fitted inside the training window of each fold.

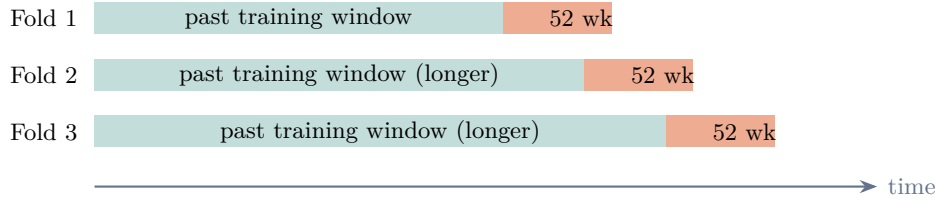


Figure 6: Expanding folds: the model never sees data later than the period it is scored on.

5.2 Information availability and forecasting assumptions

Let t denote the target week and $\mathcal{I}_t$ denote the information available when the prediction is issued.

A valid prospective forecasting pipeline must construct every predictor from information available at that time:

$$\hat{y}_t = f(\mathcal{I}_t). \quad (3)$$

In the DengAI benchmark, weather features are provided for the prediction period. Our competition experiments use these supplied climate measurements.

However, access to climate observations for the target week is not equivalent to having those measurements available before the week begins.

Therefore, the benchmark performance should be interpreted as climate-conditioned case prediction unless the operational availability of the weather measurements is independently established.

A prospective forecasting evaluation would require a defined forecast issuance time and appropriate historical observations or weather forecasts available at that time.

5.3 Metric

We use MAE because it is the competition metric and is less dominated by single extreme weeks than squared error. We report three numbers per configuration: the *mean* MAE across folds, the *worst-fold* MAE (stability), and the *outbreak-week* MAE (the weeks that matter most for public health). The same folds are used for every model so that differences measure the model and not the split.

$$\text{MAE}_{\text{outbreak},c} = \frac{1}{|\mathcal{O}_c|} \sum_{t \in \mathcal{O}_c} |y_t - \hat{y}_t|, \quad (4)$$

where $\mathcal{O}_c$ denotes the set of validation weeks classified as outbreaks for city c .

The outbreak threshold is defined using the training data and held fixed when evaluating the corresponding validation fold.

5.4 Baselines

Figure 7 compares a historical-mean predictor, a week-of-year climatology (the median or mean for that calendar week) and a tuned tree ensemble on the same 312 weeks per city. Climatology is very hard to beat in Iquitos (6.8 vs. 7.1 for the trees), whereas in San Juan climate features cut the error from 25.1 to 15.7. Figure 8 shows why San Juan is difficult: climatology predicts a smooth seasonal hump every year, while actual outbreaks are irregular.

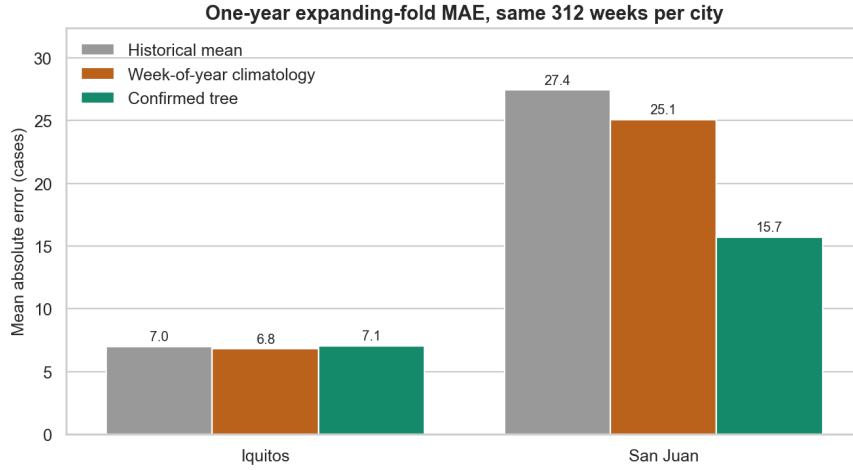


Figure 7: One-year expanding-fold MAE on identical 312 validation weeks per city.

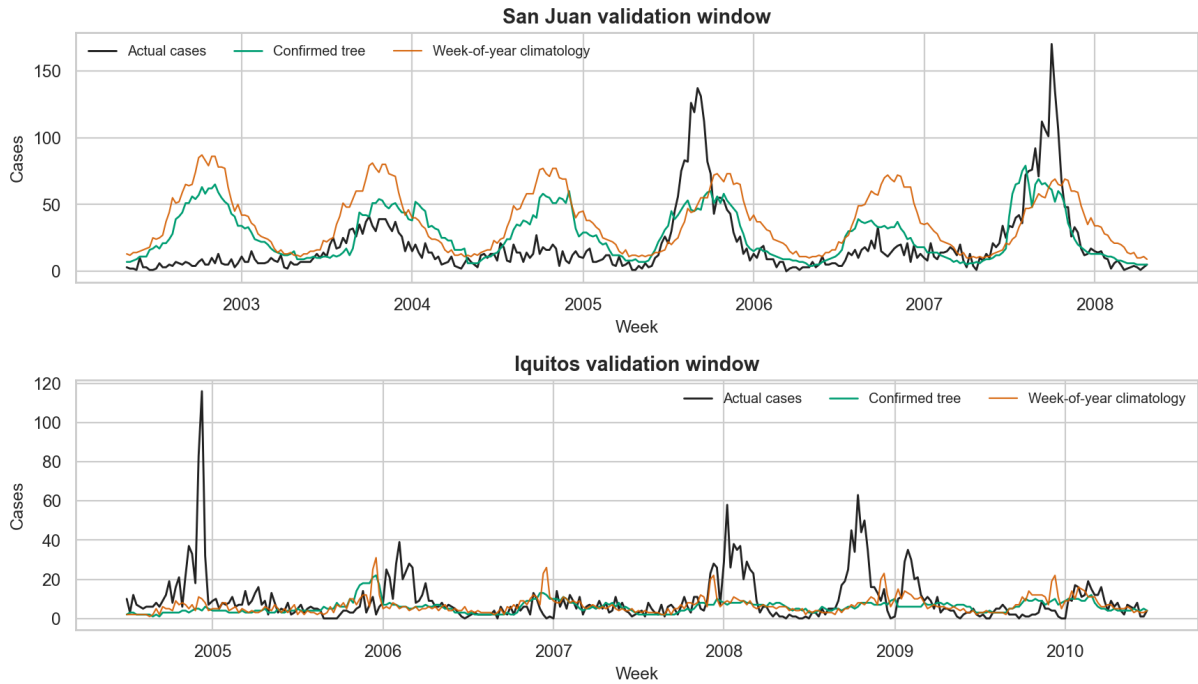


Figure 8: Validation windows: actual cases, a confirmed tree model, and week-of-year climatology. Peaks of the largest outbreaks are underestimated.

6 Models and experiments

6.1 Stage 0: the first ranking reversal

Our first serious comparison used XGBoost [2], CatBoost [3] and a Random-Forest family baseline [4]. The early training and validation runs favoured the boosted learners: sequentially correcting residuals let them fit small irregularities that a bagged forest averaged away. The archived controlled experiment confirms the local part of that observation. It selected a 20% XGBoost contribution for Iquitos and a 40% CatBoost contribution for San Juan, and both the four-fold selection score and the six-fold report score improved over the forest system (Table 4).

Table 4: The local ranking that did not survive the hidden period. The boosted row is a city-specific blend, not a claim that one boosted model won both cities.

Candidate	4-fold local MAE	6-fold local MAE	Hidden-test outcome
City-specific forest	10.091	11.393	22.5; retained
Selected boosted blend	9.788	11.061	Worse than forest; rejected

The leaderboard reversed the local ranking: the simpler forest submission generalized better. We cannot prove whether this was ordinary model-selection overfitting, temporal distribution shift, or both, because the hidden labels are unavailable. The important engineering conclusion was operational: a small local MAE advantage on a few dependent time folds was not enough evidence to trust a more adaptive model. This changed our selection rule from “take the lowest validation mean” to “also inspect fold variance, peak amplitude and behaviour on a test-length horizon.”

6.2 Stage 1: lower-variance forest ensembles (public leaderboard MAE 22.5)

We next focused on variance. Random Forest fits trees to bootstrap samples and random feature subsets. Extra Trees goes further: for candidate features it randomizes the split thresholds before choosing a split. This extra randomisation makes the individual trees less similar. If T trees have variance σ^2 and average pairwise correlation ρ , the variance of their average is approximately

$$\text{Var}(\bar{f}) = \frac{\sigma^2}{T} [1 + (T - 1)\rho]. \quad (5)$$

With $T = 500$, reducing correlation ρ is at least as important as adding more trees. Extra Trees can therefore reduce ensemble variance, although excessive randomisation can increase bias. We controlled the other major source of variance by requiring at least five training examples in every leaf.

The selected city-specific system used the causal tree features of set (A):

- San Juan: 500-tree Extra Trees [5] with Poisson splitting;
- Iquitos: 500-tree Random Forest trained on $\log(1 + y)$;
- minimum five samples per leaf to avoid very specific rules.

How the ensemble prediction is formed. This is not a vague vote among model names. Within one city, every tree produces a real-valued case forecast and the forest averages the 500 outputs:

$$\hat{y}_{c,t} = \frac{1}{500} \sum_{b=1}^{500} f_{c,b}(x_{c,t}). \quad (6)$$

For Iquitos the average is formed on the $\log(1 + y)$ scale and then inverted; for San Juan the trees predict the count directly with the Poisson split criterion. Predictions are clipped at zero and rounded to an integer. We do *not* average San Juan and Iquitos models together: the San Juan model predicts San Juan rows, the Iquitos model predicts Iquitos rows, and the two blocks are concatenated in the official submission order.

We also tested explicit forest–boosting blends,

$$\hat{y} = (1 - w)\hat{y}_{\text{forest}} + w\hat{y}_{\text{boost}}, \quad (7)$$

with $w \in \{0, 0.1, \dots, 1.0\}$. Local validation selected $w = 0.2$ with XGBoost for Iquitos and $w = 0.4$ with CatBoost for San Juan. Those blends are the second row of Table 4; because their public leaderboard result was worse, they were experiments, not the submitted 22.5 tree system. Trees capture non-linear thresholds (“warm *and* humid after several wet weeks”), lags express delayed effects, rolling means express accumulated exposure, and Fourier terms express season. This reduced public leaderboard MAE from 26.7 to 22.5. But the validation forecast in Figure 8 exposed the next problem: the forest followed the seasonal level while repeatedly cutting off the largest peaks.

Residual Analysis & Feature Engineering

We inspected the largest absolute residuals week by week. Several outbreak weeks had a sudden increase in cases even though the weather in that same row was not unusually extreme. The model had learned an overly instantaneous mapping—“this week’s weather predicts this week’s cases”—while the real process includes mosquito development, viral incubation and reporting delays. A rainfall or humidity episode can therefore matter weeks before the observed case peak.

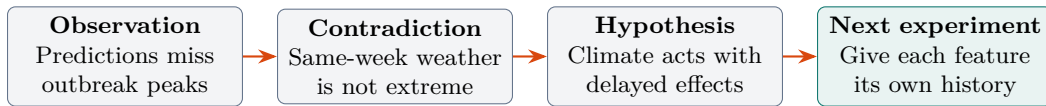


Figure 9: Error analysis converted a visible failure mode into the lag-window experiment.

This was the project’s main pivot. We stopped asking only “which learner wins?” and asked “what information is missing from the input?” The answer was not one universal lag for all variables, but a city- and feature-specific history.

6.3 Stage 2: an MLP with lagged inputs (public leaderboard MAE 19.3 → 18.8)

Lagged climate associations. We first examined the association between individual historical climate measurements and dengue incidence at different temporal lags.

Figure 10 illustrates these associations for selected weather variables. The correlations vary with lag, variable, and city, motivating further investigation of historical climate representations.

However, lagged correlation is not equivalent to predictive performance, and the lag with the largest correlation does not necessarily define the optimal history-window length.

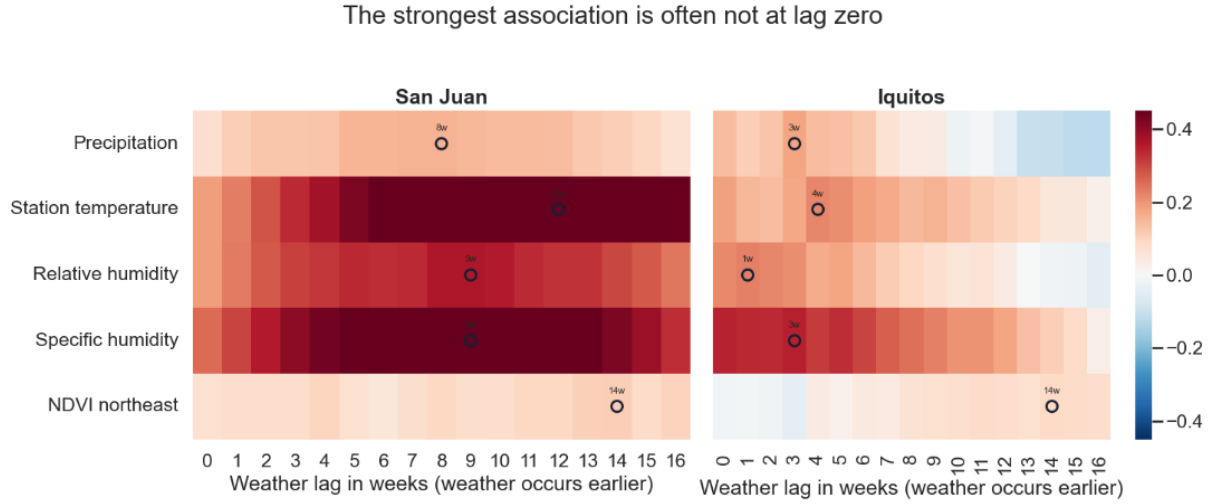


Figure 10: Lagged correlations between selected climate variables and weekly dengue cases in San Juan and Iquitos. The color scale represents signed correlation coefficients rather than prediction MAE. The plot illustrates temporal associations but does not determine the optimal length of a predictive history window.

A 40-week choice therefore means all 40 consecutive values, not only lag 40.

Candidate history lengths were selected using a separate decision-tree screening procedure evaluated by MAE.

The correlation heatmap provides exploratory motivation but should not be interpreted as the window-selection objective. For example, relative humidity used 34 weeks in San Juan and 7 in Iquitos. Because the original screen used ordinary ten-fold validation and nearby windows were often nearly tied, we treat exact values such as 39 versus 40 as engineering choices, not biological constants. The reliable insight was that climate memory is long, variable-specific and city-specific.

We then gave a neural network this structured memory. Concatenating the selected windows produced 461 inputs for San Juan and 380 for Iquitos. The network is a three-layer perceptron with SELU activations [6], dropout, a linear output and an absolute-error loss, trained with RMSprop and a plateau learning-rate decay. Figure 11 shows the two city architectures.

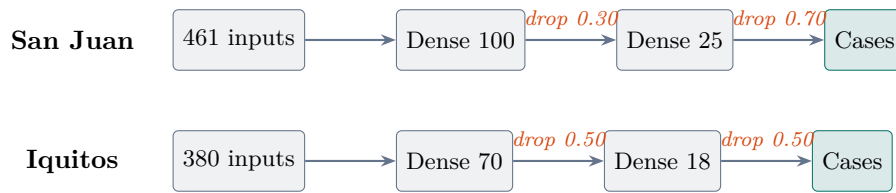


Figure 11: Raw-history MLPs after temporal tuning. In the multiscale model the input is 180 for both cities and the San Juan network keeps the same $100 \rightarrow 25$ hidden layers.

Table 5: Consecutive history retained by the raw-lag MLP. The entries are window lengths in weeks, not isolated lag positions.

Feature	SJ	IQ	Feature	SJ	IQ
Precipitation amount	40	33	Satellite precipitation	40	33
Reanalysis air temperature	16	10	Specific humidity	14	26
Reanalysis average temperature	15	4	Reanalysis diurnal range	21	34
Reanalysis dew point	39	6	Station average temperature	41	40
Reanalysis maximum temperature	12	41	Station diurnal range	40	26
Reanalysis minimum temperature	21	40	Station maximum temperature	37	39
Reanalysis precipitation	30	3	Station minimum temperature	26	25
Relative humidity	34	7	Station precipitation	32	10
Week of year	3	3			

Tuning. Under temporal cross-validation, the useful change was a *narrower second hidden layer*, not a deeper network:

Table 6: Effect of tuning the second hidden layer (mean temporal-CV MAE).

	Second layer	Before	After
San Juan	50 → 25	22.02	20.64
Iquitos	35 → 18	8.17	7.91

This gave a public leaderboard MAE of 18.8 for seed 42; averaged over three seeds it is 19.5, a reminder that a single seed overstates performance.

Remaining problem. The MLP improved the public leaderboard performance, supporting the time-lag hypothesis, but it created a statistical mismatch. The full dataset has only 1456 labels, and separate-city training leaves 936 San Juan labels and 520 Iquitos labels. Against those sample sizes, the network receives 461 and 380 highly correlated coordinates per week. In San Juan the first layer alone has 46,100 input weights. The model must discover simple concepts such as “recent mean”, “variability” and “trend” from individual positions, and shifting a wet episode by one week moves its information into many different coordinates. The representation was informative but inefficient.

Why eleven summaries per weather variable. We did not choose eleven as a magic number. We listed the distinct questions the raw trajectory should answer while remaining small: What is the latest level? What has the average level been over short, monthly, two-month, quarterly, half-year and annual horizons? How volatile has it been recently and over a quarter? Is the recent month above the quarterly baseline? Is the variable rising or falling? These questions become one current value, six means (2, 4, 8, 13, 26, 52 weeks), two standard deviations (4, 13 weeks), one short-minus-medium contrast, and one 13-week slope: $1 + 6 + 2 + 1 + 1 = 11$. The horizons align with interpretable calendar scales and overlap deliberately, allowing the MLP to compare short- and long-term exposure without seeing all 52 raw positions.

6.4 Stage 3: multiscale summaries (public leaderboard MAE 16.6)

We therefore replaced raw coordinates with the 180 summaries of feature set (C) and kept *the same MLP, training procedure and seed*. Under identical chronological folds (three folds, three seeds), summaries lower the mean and worst-fold errors in both cities (Table 7). San Juan

improves on mean, worst and outbreak error; Iquitos improves on mean and worst error, while its outbreak MAE rises slightly (from 23.2 to 23.8).

Table 7: Same tuned MLP, chronological validation, three folds and three seeds: raw lag windows vs. multiscale summaries (lower is better).

City	Representation	Mean MAE	Worst MAE	Outbreak MAE
SJ	Raw lags	20.635	31.942	57.095
SJ	Multiscale summaries	17.726	27.346	55.844
IQ	Raw lags	7.908	12.481	23.244
IQ	Multiscale summaries	7.588	10.596	23.795

The multiscale representation reduced the mean chronological-validation MAE from 20.635 to 17.726 in San Juan and from 7.908 to 7.588 in Iquitos.

On the public leaderboard, the combined MAE improved from 18.8 to 16.6.

Is it compression or just more information?

If the summaries helped by adding information, then raw lags *plus* summaries should be best. In a two-fold ablation it was the worst (Figure 12): San Juan MAE was 17.86 for raw histories, **12.61** for summaries and 18.43 for raw plus summaries. These results suggest that replacing highly correlated raw lag coordinates with compact multiscale summaries provides a more useful inductive bias for the MLP.

However, the experiment does not isolate the effects of dimensionality reduction, optimization, and regularization.

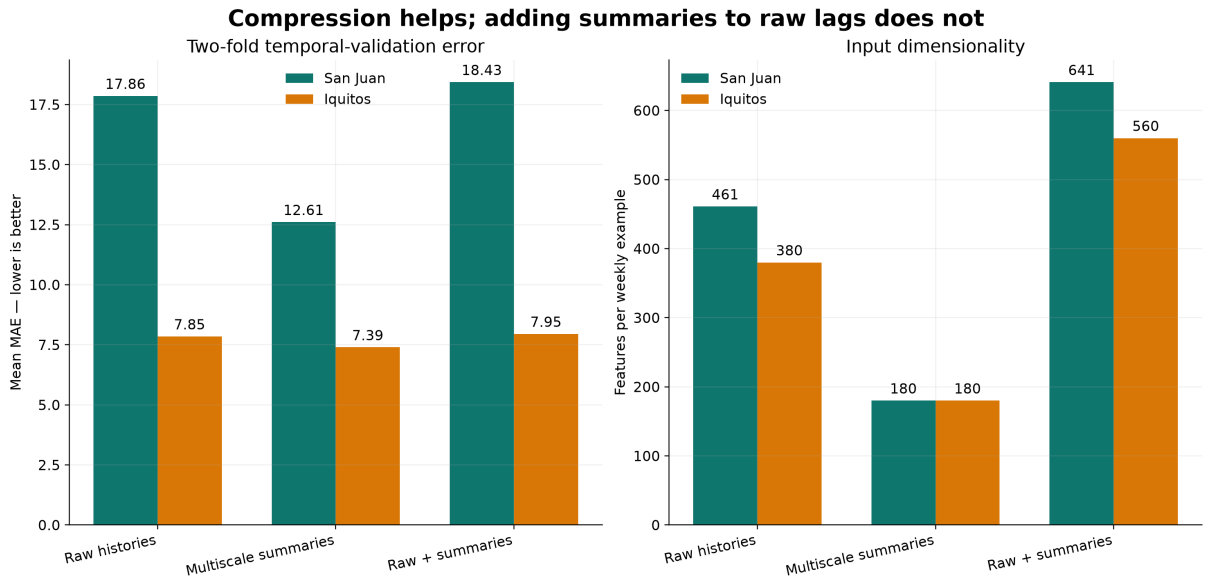


Figure 12: Left: two-fold temporal-validation MAE for three input representations. Right: number of inputs per weekly example. Compression helps; adding summaries to raw lags does not.

6.5 Stage 4: select, round and reproduce the exact 15.9832 submission

The restored scored file resolved an important provenance question. Its recorded recipe contains no PLS transformation. Instead, it combines the best confirmed pipeline for each city:

Table 8: Hash-verified recipe of the final submission.

City	Representation	MLP	Selection
San Juan	180 multiscale inputs ($16 \times 11 + 4$ season terms)	$180 \rightarrow 100 \rightarrow 25 \rightarrow 1$, seed 42	nearest-integer rounding
Iquitos	380 feature-specific raw-history inputs	$380 \rightarrow 70 \rightarrow 18 \rightarrow 1$, seed 42	preserved raw-lag baseline

San Juan uses SELU hidden units, dropout 0.3/0.7, learning rate 0.01, mini-batches of 16, and 40×200 update steps. After clipping to non-negative values, its continuous output is converted by

$$\hat{y}_{\text{int}} = \lfloor \max(\hat{y}, 0) + 0.5 \rfloor. \quad (8)$$

This rule raised 128 of the 260 San Juan test predictions by one case relative to truncation and improved the chronological hold-out MAE from 16.635 to 16.519. Iquitos remains the archived seed-42 raw-lag prediction because multiscale compression did not improve that city’s outbreak behaviour consistently.

Reproduction contract. The notebook pins Python 3.12, NumPy 2.4.4 and pandas 3.0.2, recreates the submission, and compares it with the restored reference. The result is exact equality in all 416 rows and the same SHA-256 digest:

b276c071224e0dab0bf9a875bb5496107de28c7c539c0bcd64aa4465fc08b8fd.

This version pin matters: the same code under an older NumPy environment produced numerically different predictions. The leaderboard labels are private, so the notebook cannot recompute 15.9832 locally; it establishes that the reproduced CSV is byte-identical to the file associated with that observed score.

Provenance correction. PLS was investigated later as a supervised compression experiment [7], but it is not present in this hash-verified submission. An earlier report draft incorrectly attributed the final leaderboard improvement to PLS. The final score must therefore be described as the result of the city-specific hybrid, the selected San Juan seed and rounding rule, not as a PLS gain.

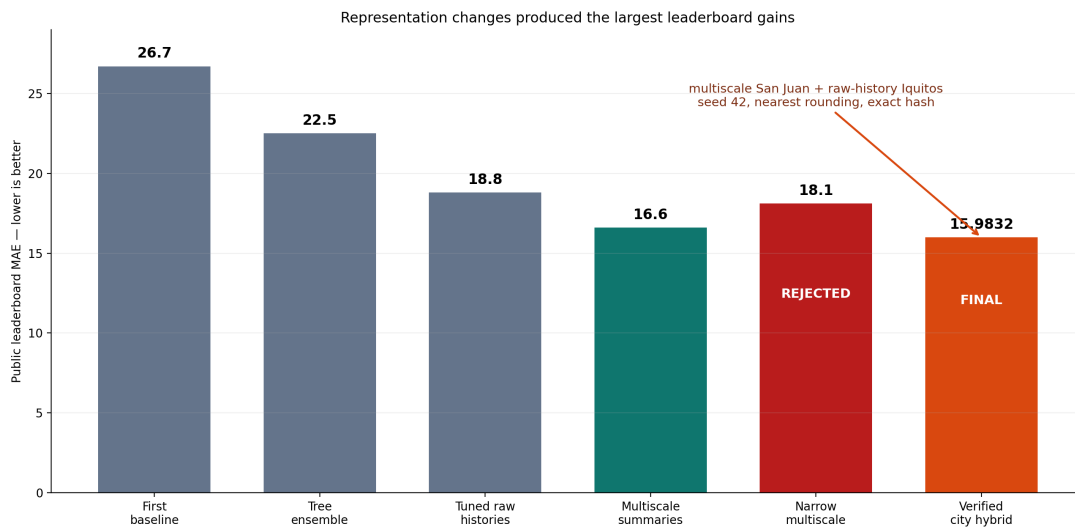
7 Results and analysis

7.1 The leaderboard journey

Table 9 and Figure 13 summarise the public leaderboard MAE of every stage.

Table 9: Observed public leaderboard MAE across stages (lower is better).

Stage	Representation	Predictor	MAE
First baseline	Earlier feature pipeline	Baseline model	26.7
Tree system	Current climate, sparse lags, rolling means, season	SJ Extra Trees; IQ Random Forest	22.5
Raw-history MLP	Feature-specific contiguous windows	3-layer MLP	19.3
Improved dropout	Same windows	3-layer MLP	19.1
Tuned raw-history MLP	461 SJ / 380 IQ raw values	Tuned MLP, seed 42	18.8
Multiscale MLP	180 summaries	Same tuned MLP	16.6
Narrow multiscale	Same 180 summaries	48 $\rightarrow$ 12 MLP	18.1 (rejected)
Verified city hybrid	SJ: 180 summaries; IQ: 380 raw lags	Seed 42 + nearest rounding	15.9832

**Figure 13:** Observed public leaderboard MAE by stage. The final bar is the hash-verified city hybrid: multiscale San Juan, raw-history Iquitos and nearest-integer rounding.

7.2 The rejected experiments were part of the result

The score sequence can look deceptively direct in hindsight. In practice, much of the work was spent establishing what *not* to trust. We preserved the most informative negative results in Table 10; they explain why the final system is a hybrid rather than the winner of one large hyperparameter search.

Table 10: Experiments that did not enter the final submission and the information they contributed.

Experiment	What happened	Lesson carried forward
XGBoost/CatBoost and forest blends	Better training/development behaviour and lower local MAE, but worse hidden generalisation	Temporal validation is evidence, not proof; preserve a simple fallback and check prediction amplitude
Forest hyperparameter and component search	Several mixtures won on development origins but not on the untouched recent origin	Prefer small, stable gains; use test-length chronological horizons
Negative-binomial, volatility, recency and quantile corrections	Plausible count/time-series ideas did not improve robustly enough to replace the confirmed forest	A sophisticated output model cannot recover climate history that is absent from the input
Three-seed raw-lag MLP ensemble	Averaging seeds scored 19.5, while seed 42 scored 18.8	Averaging highly correlated models can shrink rare peaks instead of adding useful diversity
Raw lags plus summaries	More inputs were worse than summaries alone (18.43 vs. 12.61 in the two-fold ablation)	The gain was compression and inductive bias, not simply more information
Narrow 48 $\rightarrow$ 12 multiscale MLP	Won local architecture search by 0.224 MAE but lost public leaderboard by 1.5 MAE	Freeze the proven network when a searched local gain is smaller than fold variance

This effort was worthwhile because each negative result narrowed the diagnosis. Model-family tuning exposed generalisation risk; peak-by-peak residual inspection exposed missing temporal context; the raw-plus-summary ablation provided additional evidence that compact temporal representations were more effective than simply increasing the number of inputs under the tested model configuration. The final score is therefore the end of an evidence chain, not a lucky architecture choice.

7.3 Local validation is evidence, not proof

After shrinking the input from 461 to 180, a local architecture search preferred a narrower 48 $\rightarrow$ 12 network by only 0.224 validation MAE, and its outbreak error was already worse. On the public leaderboard it was 1.5 MAE *worse* (18.1 vs. 16.6; Figure 14). A small win after searching many candidates is likely selection noise. The representation survived public leaderboard testing; the architecture difference did not. We therefore froze the 100 $\rightarrow$ 25 San Juan network and concentrated on controlled ablations of the representation.

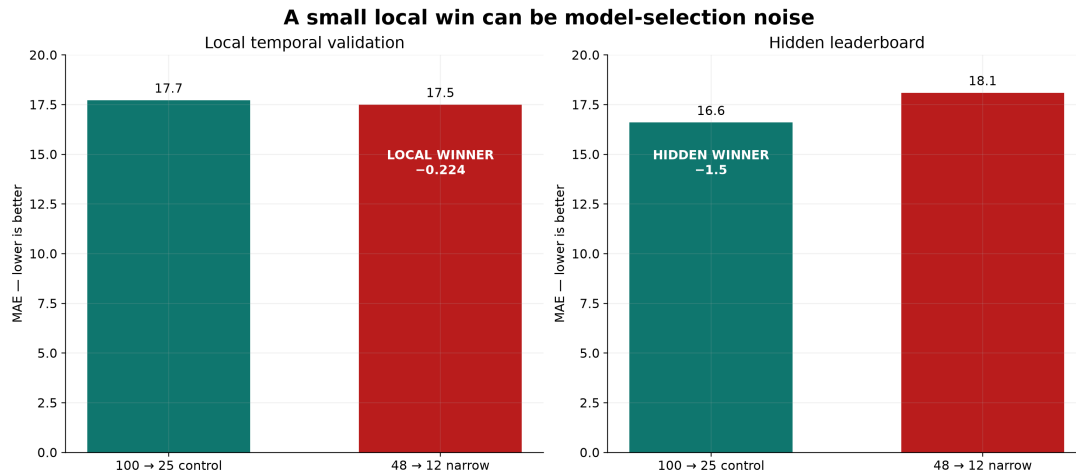


Figure 14: The 48 → 12 network looked better locally (−0.224) but was worse on the public leaderboard (+1.5).

7.4 Final submission and leaderboard performance

The final submission combines city-specific models with different temporal representations.

For San Juan, we use 180 multiscale climate features. This representation is processed by an MLP with hidden layers of 100 and 25 units.

For Iquitos, we retain the tuned raw-history MLP with 380 input features and hidden layers of 70 and 18 units.

Both models generate non-negative integer predictions, which are assembled according to the official submission template. San Juan uses nearest-integer rounding; Iquitos predictions come from the preserved seed-42 raw-history file.

Submissions (9)

- To track competition progress, submissions are scored against public test data to give a "public score".
- The primary evaluation metric is Mean Absolute Error. [More](#)

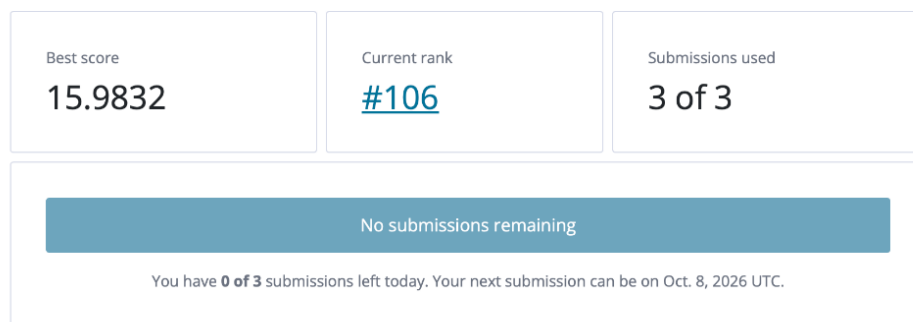


Figure 15: Final DengAI public leaderboard result. The model achieved an MAE of 15.9832 and was ranked 106th at the time of capture. This result is distinct from chronological cross-validation performance.

The final system achieved a public leaderboard MAE of **15.9832**, with a rank of **106** at the time of evaluation, as shown in Figure 15.

This result provides evidence of performance on the competition's public evaluation period. However, it does not independently establish generalization to unseen cities or future epidemic periods.

Detailed implementation instructions, experimental configurations, and source code are provided in our GitHub repository.

The reproduction notebook recreated all 416 predictions exactly. Both the restored and generated files have the following SHA-256:

b276c071224e0dab0bf9a875bb5496107de28c7c539c0bcd64aa4465fc08b8fd.

7.5 Error analysis

- **Outbreaks changed the representation.** In San Juan the outbreak-week MAE (≈ 56) is more than three times the mean MAE (≈ 18) (Table 7). The early trees predicted the seasonal envelope but missed abrupt peaks whose same-week weather looked ordinary. That residual pattern, not a generic desire for more features, motivated delayed climate windows and then multiscale summaries. The final model times major rises more convincingly, although peak amplitude remains uncertain.
- **Fold variance is large.** The worst San Juan fold (27.3) is much worse than the mean (17.7): performance depends on which year is held out.
- **Single seeds mislead.** The 18.8 result had a three-seed mean of 19.5.
- **Iquitos is near its floor.** Climatology is as good as complex models on Iquitos validation weeks (≈ 7), consistent with its small, noisy counts.

8 Discussion

8.1 Feature engineering as the primary driver of performance

The central finding of this project is that the representation of climate history played a more important role in our experiments than increasing model complexity. Rather than continuously introducing more sophisticated predictive models, our largest improvements came from investigating what information the models received and how that information was represented.

From instantaneous observations to temporal history. Our initial tree-based models used current climate measurements, selected lags, rolling statistics, and seasonal features. Although these models captured general seasonal patterns, error analysis revealed that they frequently underestimated large dengue outbreaks.

This observation motivated us to investigate delayed relationships between climate conditions and dengue incidence. We constructed city- and variable-specific history windows, producing 461 input coordinates for San Juan and 380 for Iquitos.

The raw-history MLP improved the public leaderboard MAE from 22.5 to 18.8 after temporal and architectural tuning. This result suggested that richer historical climate information could be useful for prediction. However, because both the model family and feature representation changed, this improvement cannot be attributed exclusively to feature engineering.

From raw histories to multiscale representations. Although raw histories preserve detailed temporal information, they also introduce hundreds of highly correlated inputs. This creates an inefficient representation for a dataset containing only 936 San Juan and 520 Iquitos training observations.

We addressed this limitation by constructing 11 statistics for each of the 16 weather variables, together with four cyclic seasonal features, resulting in 180-dimensional input vectors.

These features represent climate conditions at multiple timescales, including current levels, short- and long-term averages, variability, recent changes, and temporal trends.

Importantly, the multiscale representation does not merely reduce dimensionality. It introduces an inductive bias by explicitly providing temporal patterns that the MLP would otherwise need to learn from individual weekly observations.

Under the same tuned MLP and chronological validation procedure, the multiscale representation reduced the mean MAE in San Juan from 20.635 to 17.726 and in Iquitos from 7.908 to 7.588.

The public leaderboard MAE also improved from 18.8 to 16.6.

These controlled experiments provide our strongest evidence that temporal feature representation was a major contributor to predictive performance.

Why adding more features was not sufficient. We additionally compared raw histories, multiscale summaries, and their concatenation.

The combined representation performed worse than the multiscale representation alone in the reported two-fold ablation.

This finding suggests that useful feature engineering is not simply the addition of more information or input dimensions.

Instead, constructing compact and meaningful representations can improve the inductive bias and optimization behavior of neural networks trained on relatively small datasets.

Nevertheless, this experiment does not independently isolate dimensionality reduction from changes in optimization and regularization.

Artifact recovery changed the conclusion. The final CSV is part of the experimental evidence, not merely an export. Recovering it allowed us to match every prediction to its recorded recipe and correct a mistaken PLS attribution. The notebook also showed that the numerical environment is part of the model: Python 3.12 with NumPy 2.4.4 reproduced the artifact exactly, whereas an older NumPy environment did not. For small neural models, code, data, package versions, output file and hash must therefore be archived together.

8.2 Limitations and implications

Despite these improvements, dengue prediction remains challenging, particularly during rare and extreme outbreaks.

Our experiments showed that lower overall MAE does not necessarily imply better prediction of outbreak weeks. In Iquitos, for example, multiscale summaries slightly increased outbreak-week MAE despite improving average validation performance.

Furthermore, our results are limited to two cities, a relatively small historical dataset, and the climate measurements supplied by the competition.

The historical feature-construction pipeline also contains alignment and temporal-context behaviors that require further causal auditing before prospective forecasting claims can be established.

Therefore, the observed leaderboard performance should not be interpreted as direct evidence of real-world early-warning effectiveness.

Main lesson. Our findings emphasize that, particularly for small time-series datasets, the quality and structure of input features can be as important as the choice of predictive algorithm.

In our experiments, the most convincing improvements emerged from diagnosing missing temporal information, constructing multiscale summaries, and preserving a city-specific hybrid.

The broader lesson is that feature engineering should be treated as a central part of model design rather than merely a preprocessing step.

9 Future work

- Repeat the multiscale MLP over more temporal cut points and seeds, and archive each exact submitted CSV, manifest and hash alongside its score.
- Evaluate PLS prospectively as a separate leakage-safe experiment rather than attributing it to the existing 15.9832 artifact.
- Quantify uncertainty (quantile loss or ensembles over seeds) to give outbreak risk bands rather than a single count.
- Evaluate more folds and repeated seeds before accepting architecture changes smaller than the fold-to-fold spread.
- Test the representation on other cities or other vector-borne diseases.

10 Conclusion

This project investigated weekly dengue case prediction for San Juan and Iquitos using climate and vegetation measurements. Rather than treating model selection as the primary challenge, we progressively examined how temporal dependencies, feature representations, and model complexity influence predictive performance.

Our experiments produced three main findings. First, a model with better chronological validation performance did not necessarily achieve a better public leaderboard result, demonstrating the limitations of selecting models using a small number of temporal folds. Second, representing climate history through multiscale statistical summaries improved the performance of a fixed MLP architecture compared with raw lagged inputs. Third, the strongest submission was a city-specific hybrid rather than a single universal representation: San Juan used multiscale summaries and nearest rounding, while Iquitos retained its raw-history model.

The final hybrid system achieved a public leaderboard MAE of **15.9832**. Its 416 predictions were reproduced exactly with a pinned environment and verified by SHA-256. Nevertheless, outbreak prediction remains difficult, and the benchmark does not directly establish prospective forecasting performance. Future work should examine strictly causal preprocessing, repeated-seed stability, and evaluation under realistic information-availability constraints.

Overall, the project highlights the importance of diagnosing model errors and designing informative temporal representations, rather than relying exclusively on increasingly complex predictive architectures.

Author Contributions

- **Duy Chien Nguyen:** Conducted data preprocessing, developed the baseline models, investigated Partial Least Squares (PLS) as a later representation experiment, made the presentation slides, and wrote the report.
- **Dang Quy Nguyen:** Conducted data preprocessing, developed and improved the tree-based methodologies, led feature engineering efforts, built the lagged feature representations

and the Multi-Layer Perceptron (MLP) architecture, made the presentation slides, and wrote the report.

Code Availability

- Data: DengAI training/test CSV files from the competition site.
- **Code Repository:** The complete source code, trained models, and reproducible notebooks are available on GitHub at <https://github.com/dangnq2501/DengAI>.
- **Presentation Video:** A recorded video of our project presentation can be viewed at https://youtu.be/Et3oHyoI_VI.
- The public leaderboard MAE 15.9832 is a leaderboard observation associated with the restored artifact; hidden labels are unavailable locally.
- Reproduction generated the same 416 predictions. Its SHA-256 is:
b276c071224e0dab0bf9a875bb5496107de28c7c539c0bcd64aa4465fc08b8fd.

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